



Pediatric MRI Simulator: User Manual

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1.0 Introduction

The MRI Simulator Employee Manual provides hospital staff with step-by-step instructions for operating a Simulator's software system. This includes controlling lights, sound, bed movement, and emergency functions that support a safe and realistic MRI experience for pediatric patients.

The purpose of the manual is to ensure smooth operation, promote child safety and maintain consistency across all sessions. In order to comply with Fraser Health's operating and safety regulations, it also describes the necessary steps for setup, customisation, cleaning, troubleshooting, and system shutdown.

This manual is meant for clinical staff, child life experts, and authorized workers who use the MRI simulator during anxiety-reduction or patient preparation sessions.

2.0 System Overview

3.0 Getting Started (Powering On, Initial System Checks)

The MRI Simulator uses a touchscreen interface that controls lighting, audio, and cycle execution. Once the screen turns on the user is prompted to SignIn with the password. The screen password has been added to ensure safety, and security. This prohibits non-fraser health employees to access the system that may cause any system damages or risk pediatric patient's safety.



Enter Pin

Enter

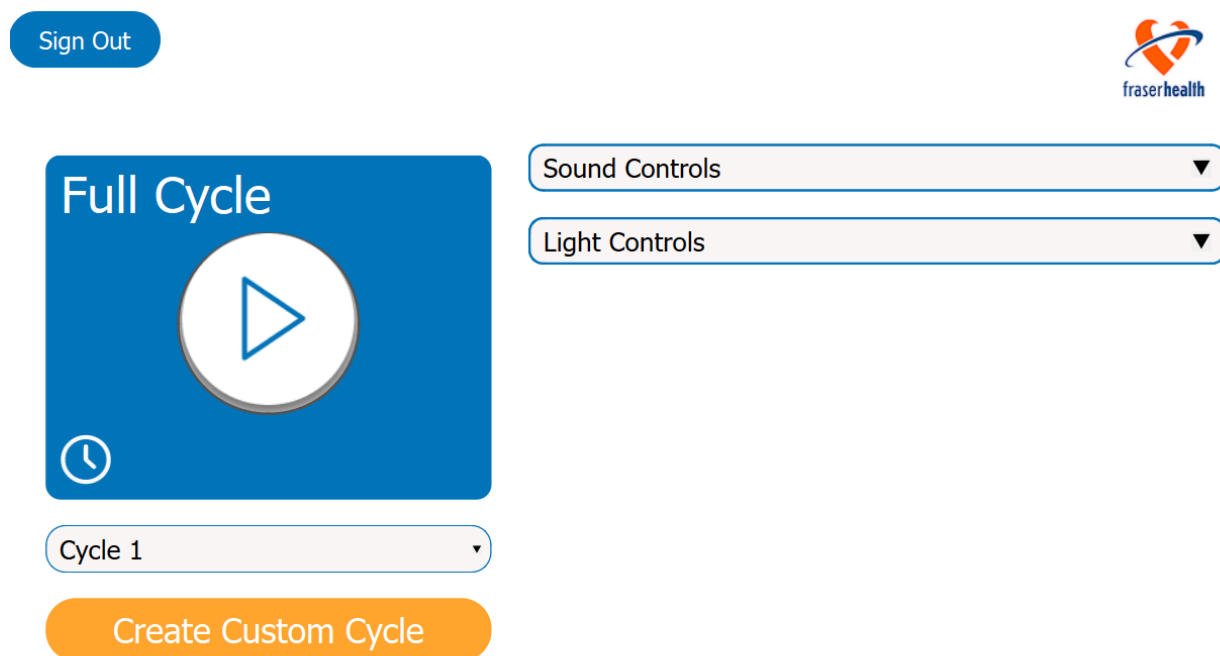
1	2	3
4	5	6
7	8	9
del	0	clr

The pin has been set to **2026**. The user can use the pin pad on the right of the screen to enter the pin and click “Enter” once done. If the pin is correct the user is navigated to the home screen. If a pin other than **2026** is entered, the user is exempted from signing in. The user has an allowance of 3 attempts to enter the pin. If the user reaches the upper limit of attempts then the screen timeout for 15secs, prohibiting the usage of the screen. After timeout, the user can try again with the valid pin.

4.0 Home Page Navigation

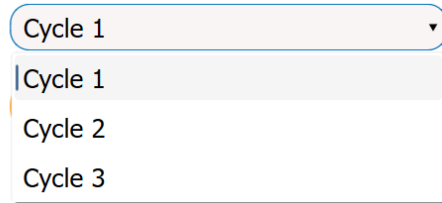
After signing in securely, the user can see the home screen. The full cycle refers to the preset cycle. The dropdown menu below it, allows the user to select from the preset cycle. The 3 set preset cycles have different sound combinations, light controls and total length. These cycles are not customisable. If the patient would like to customize the cycle, the employee in duty is encouraged to customize the cycle for the patient by following the guidelines discussed in the following section.

Fig1: Home Screen



4.1 Preset Cycles

In order to select a cycle the user can click the dropdown menu and choose from 3 preset cycles.



Once the cycle is selected users have access to review the preset sound and brightness level for each cycle. As a reference cycle 1 is selected and is highlighted.

Each cycle has following sounds:

1. mri_sound1: Pulse
2. mri_sound2: Drill(S)
3. mri_sound3: Drill(M)
4. mri_sound4: Beeps
5. mri_sound5: Scan
6. mri_sound6: Alarm
7. mri_sound7: Buzz
8. mri_sound8: Drill(L)

Each Cycle have varied running time:

1. Cycle 1: 1 min
2. Cycle 2: 3mins
3. Cycle 3: 5mins

5.0 Create Custom Cycle

6.0 System Shutdown Procedure

7.0 Troubleshooting

8.0 Safety Notes

9.0 Contact and Support